AISWARYA JAYACHANDRAN

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SUMMARY

M.S. Computer Science student at Georgia Tech with 3 years of C/C++ systems engineering experience across large-scale, performance-sensitive software. Skilled in building RESTful APIs, interactive UIs and full-stack applications, with hands-on work in distributed data processing using Spark and Databricks, Docker and Linux-based development. Strong foundation in data structures, algorithms and software design.

EDUCATION

Georgia Institute of Technology — **May 2027 (Expected)**
Master of Science in Computer Science, Computing Systems **(4.0/4.0)** — Atlanta, USA

Govt. Model Engineering College — **June 2022**
Bachelor of Technology in Computer Science and Engineering **(9.1/10)** — Kerala, India

Related Coursework: Data Structures and Algorithms, Computer Networks, Machine Learning, Data Visualization & Analytics High Performance Computer Architecture

TECHNICAL SKILLS

Programming Languages	C/C++, JavaScript, TypeScript, Python, TCL
Frontend	React, HTML5, CSS, Bootstrap, Chart.js, Plotly
Backend & APIs	Node.js, Express.js, Django, REST APIs, API integration
Databases	PostgreSQL, MongoDB, SQLite, SQL
Cloud & Tools	Docker, AWS, Git, GitHub, GNU Debugger, Perforce, VMware, Visual Studio Code

PROFESSIONAL EXPERIENCE

Georgia Institute of Technology, College of Computing — **Jan 2026 - May 2026**
Graduate Teaching Assistant for Data Visualization and Analytics — Atlanta, USA

- Managed office hours and course operations for **300+** students, mentoring through assignments involving data cleaning with **OpenRefine**, SQL using **SQLite**, visualization with **D3.js** and distributed processing using **Spark**, **Databricks**, **Docker** and cloud platforms like **AWS/GCP.**

Synopsys Inc. — **Jan 2025 - Aug 2025**
Senior Research and Development Engineer — Bangalore, India

- Debugged and resolved **80+** complex **C/C++** software defects using **GNU Debugger** across large-scale EDA codebases, improving reliability of simulation and design-validation workflows.
- Drove a **key customer's tool migration** as **primary technical point of contact**, coordinating across engineering and support teams to resolve critical blockers within one week of upgrade, ensuring on-time transition and customer sign-off.
- Led the team's **GitHub Copilot** adoption initiative, defining usage guidelines and presenting adoption progress to the team, producing supporting documentation that accelerated ramp-up on AI-assisted development workflows.

Synopsys Inc. — **Jul 2022 - Dec 2024**
Research and Development Engineer — Bangalore, India

- Delivered **C/C++** features for PrimePower, an IC power-analysis tool, including power annotation functionality used in performance-sensitive design-analysis workflows.
- Built and maintained **100+ regression test suites** using **TCL** scripting covering **20+ scenarios**, ensuring reliable tool performance across releases.
- Coordinated 3 cross-functional VERDI tool-version migrations, managing engineering dependencies and presenting migration guidelines to **20+** engineers.

Aarna Analytics — **Aug 2020 - Feb 2021**
Frontend Developer Intern — Texas, USA (Remote)

- Launched the webpage for Profitis, an AI-powered prediction engine, and built a custom CRM application with a **React** and **Java Spring Boot** for home chefs to manage inventory, orders and customer workflows.

ACADEMIC PROJECTS

Longitudinal Dataset to Characterize IETF RFC Publication and Deployment — **Aug 2025 - Apr 2026**

- Primary researcher on a faculty-supervised project targeting an **ACM IMC 2026** submission, driving research design, analysis, and manuscript preparation.
- Built an automated data pipeline using **Docker** and **MongoDB** , integrating **OpenAI API** for **LLM-based** metadata cleaning and deployment-label extension validated against ground-truth labels.

Multilevel Cache Simulator | C/C++ — **Jan 2026 - Feb 2026**

- Built a **C++ trace-driven L1/L2 cache simulator** with configurable size, associativity, block size, write policies, LRU eviction and MIP/LIP insertion.
- Implemented **Markov and hybrid memory prefetchers** using transition-frequency tables, evaluating hit/miss rates and **AAT** across benchmark traces.

US Natural Hazard Risk Assessment Dashboard | Python, JavaScript, Chart.js, Leaflet — **Aug 2025 - Dec 2025**

- Created an interactive U.S. natural disaster risk dashboard using **React, JavaScript, Chart.js and Plotly,** integrating NOAA and FEMA NRI datasets with state-level heatmaps, filters and trend visualizations to compare regional hazard risk.
- Developed a **2025 disaster impact prediction pipeline** using **Transformer, XGBoost** and **Random Forest** models to forecast disaster frequency, fatalities and economic loss, with **SHAP** analysis for risk-driver interpretation.